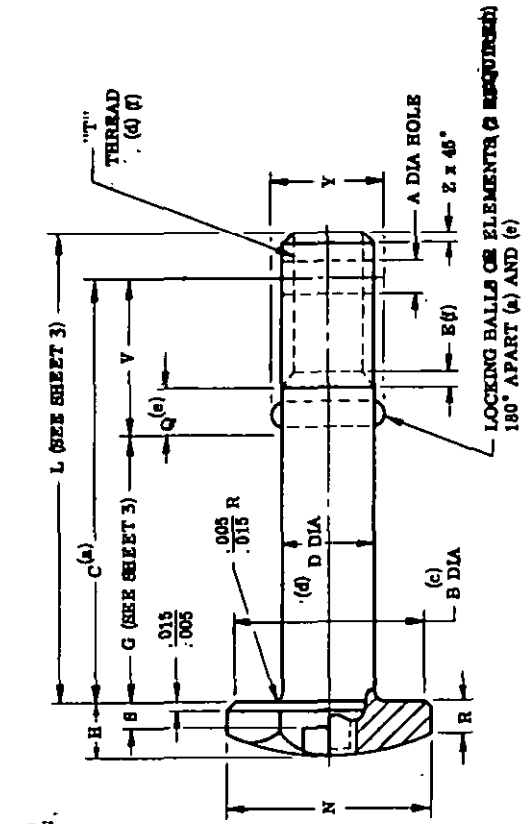
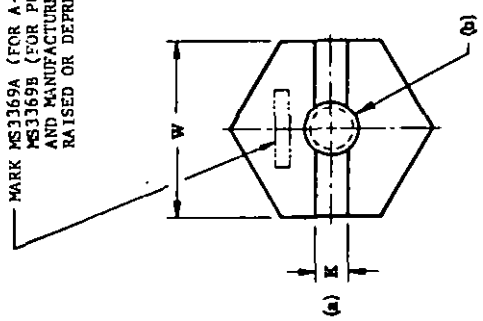
FED. SUP CLASS
5306





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APPROVED 7 APR 1969 REVISED (B) 12 OCT 79 (C) 11 MARCH 1985 (D) 20 JUN 85

(D) DENOTES CHANGE

P.A. NAVY - AS Other Cust USAF - 11 ARMY - AV	TITLE BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 63 KSI Ftu, HEXAGON SLOTTED HEAD, 450°F AND 650°F	MILITARY STANDARD MS3369 SHEET 1 OF 8
PROCUREMENT SPECIFICATION MIL-B-23964	SUPERSEDES MS3369 (AS)	

DD FORM 1 MAR 72 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. PROJECT NO. 5306-0816 PLATE NO. 23085

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PROCUREMENT SPECIFICATION: MIL-B-23964

P.A.: NAVY - AS
Other Cust: USAF-11
ARMY-AV

SUPERSEDES: MS3369 (AS)

TITLE: BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 63 KSI Fcu, HEXAGON SLOTTED HEAD, 450°F AND 650°F

MS3369

MILITARY STANDARD

TABLE I

DASH NO.	"T" THREAD DESIGNATION	A		B	D··		H		K		N	Q		R		S		V	W·		Y		Z	
		MAX	MIN	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MIN	MAX	MIN	MAX	MIN	MAX	MIN	BASIC	MAX	MIN	MAX	MIN	MAX	MIN
-3	.1900-32UNJF-3A	.080	.070	.335	.1894	.1889	.110	.095	.057	.047	.413	.094	.079	.083	.063	.032	.022	.293	.376	.367	.225	.215	.041	.021
-4	.2500-28UNJF-3A	.086	.076	.398	.2492	.2487	.141	.126	.073	.063	.488	.125	.110	.104	.084			.333	.439	.430	.294	.284		
-5	.3125-24UNJF-3A			.460	.3117	.3112	.172	.157	.088	.078	.557	.156	.141	.125	.105	.047	.037	.375	.502	.492	.380	.370		
-6	.3750-24UNJF-3A			.522	.3742	.3737	.204	.189	.104	.094	.628	.171	.156	.146	.126			.421	.564	.553	.445	.435	.067	.037
-7	.4375-20UNJF-3A	.116	.106	.585	.4367	.4362	.235	.220			.698	.202	.187	.166	.146	.063	.053	.460	.627	.617	.514	.504		
-8	.5000-20UNJF-3A			.710	.4991	.4986	.266	.251	.119	.109	.840	.234	.219	.188	.168	.078	.068	.523	.752	.741	.599	.589		
-9	.5625-18UNJF-3A	.151	.141	.835	.5616	.5611	.297	.282			.910	.265	.250	.208	.188	.094	.084	.628	.877	.865	.671	.661	.072	.052
-10	.6250-18UNJF-3A	.151	.141	.898	.6241	.6236	.328	.313	.135	.125	1.050	.305	.290	.228	.208	.125	.115	.682	.940	.928	.756	.746	.072	.052
-12	.7500-16UNJF-3A	.151	.141	1.022	.7491	.7486	.391	.376	.151	.141	1.200	.343	.328	.271	.251	.140	.130	.815	1.064	1.052	.893	.883	.087	.067

(D)

GENERAL DIMENSIONAL AND MECHANICAL REQUIREMENTS:

(a) COTTER PIN HOLE SHALL BE WITHIN .010 INCH OF BOLT CENTERLINE FOR BOLTS 5/16 DIA AND SMALLER; AND WITHIN .015 INCH FOR BOLTS 3/8 DIA AND LARGER. THE PIN HOLE SHALL BE PERPENDICULAR TO BOLT CENTERLINE WITHIN ±2°. LOCATION OF COTTER PIN HOLE FROM BEARING SURFACE IS GIVEN BY C = [G MIN + V] ±.010. THE COTTER PIN HOLE SHALL BE PERPENDICULAR TO THE SLOT IN HEAD WITHIN ±5°. ORIENTATION OF BALLS OR ELEMENT TO COTTER PIN HOLE SHALL BE IN LINE WITHIN ±3°.

(b) THE RETAINING DEVICE SHALL BE RELEASED ONLY BY DEPRESSING THE PLUNGER LOCATED IN THE HEAD OF THE BOLT.

(c) RADIUS RELIEF OR CHAMFER TO "B" DIA.

(d) CONCENTRICITY: D DIA SHALL BE CONCENTRIC TO P.D. OF THREAD WITHIN .002 TIR AT MAXIMUM MATERIAL CONDITION.

(e) BALLS OR RETAINING ELEMENTS SHALL BE LOCATED WITHIN THE LIMITS OF Q DIMENSION AND AT MAXIMUM PROTRUSION SHALL FIT WITHIN THE GROOVES OF THE APPROPRIATE SPACER OF MS21126 AND MS21128 WITHOUT INTERFERENCE OR BINDING. BALL OR ELEMENT PROTRUSION FROM THE BOLT CENTERLINE SHALL BE EQUAL TO OR LESS THAN 1/2 Y MAX.

(f) THREAD: IN ACCORDANCE WITH MIL-S-8879 EXCEPT THAT INCOMPLETE THREAD LENGTH (E) SHALL BE A MAXIMUM OF 1-1/2 AND A MINIMUM OF 1/2 THREAD PITCHES IN LENGTH AND THE MAJOR DIA SHALL BE .001 LESS THAN THE MEASURED SHANK DIA.

FED. SUP. CLASS 5306

APPROVED 7 APR 1969

REVISED (D) FOR CHANGES SEE SHEET 2

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PLATE NO 27265

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USER SYMBOLS:

DD FORM 1426/72 (Superseded)

PROCUREMENT SPECIFICATION
MIL-B-22984

P.A.
NAVY - AS
Other Cust
USAF - 11
ARMY - AV

SUPERSEDES
MS9369 (AS)

TITLE
BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Ftu, 63 KSI Fsu, HEXAGON SLOTTED HEAD, 450° F AND 650°F

MILITARY STANDARD
MS3369

SHEET 3 OF 8

FED. SUP CLASS
5306

TABLE II

GRIP DASH NO.	NOMINAL SIZE		.1900 (10)		.2500 (1/4)		.3125 (5/16)		.3750 (3/8)		.4375 (7/16)		.5000 (1/2)		.5625 (9/16)		GRIP DASH NO.
	THREAD DESIGNATION		32 UNJF-3A		28 UNJF-3A		24 UNJF-3A		24 UNJF-3A		20 UNJF-3A		20 UNJF-3A		18 UNJF-3A		
	G-GRIP LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		
	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	
03	.204	.188	.580	.560	.626	.606											03
04	.266	.250	.642	.622	.688	.668	.748	.728	.809	.789							04
05	.328	.312	.705	.685	.751	.731	.810	.790	.871	.851							05
06	.390	.375	.767	.747	.813	.793	.873	.853	.934	.914	.978	.958	1.041	1.021			06
07	.454	.438	.830	.810	.876	.856	.935	.915	.996	.976	1.040	1.020	1.103	1.083	1.244	1.224	07
08	.516	.500	.892	.872	.938	.918	.998	.978	1.059	1.039	1.103	1.083	1.166	1.176	1.306	1.286	08
09	.578	.562	.955	.935	1.001	.981	1.060	1.040	1.121	1.101	1.165	1.145	1.228	1.208	1.369	1.349	09
10	.641	.625	1.017	.997	1.063	1.043	1.123	1.103	1.184	1.164	1.228	1.208	1.291	1.271	1.431	1.411	10
11	.704	.688	1.080	1.060	1.126	1.106	1.185	1.165	1.246	1.226	1.290	1.270	1.353	1.333	1.494	1.474	11
12	.766	.750	1.142	1.122	1.188	1.168	1.248	1.228	1.309	1.289	1.353	1.333	1.416	1.396	1.556	1.536	12
13	.828	.812	1.205	1.185	1.251	1.231	1.310	1.290	1.371	1.351	1.415	1.395	1.478	1.458	1.619	1.599	13
14	.891	.875	1.267	1.247	1.313	1.293	1.373	1.353	1.434	1.414	1.478	1.458	1.541	1.521	1.681	1.661	14
15	.954	.938	1.330	1.310	1.376	1.356	1.435	1.415	1.496	1.476	1.540	1.520	1.603	1.583	1.744	1.724	15
16	1.016	1.000	1.392	1.372	1.438	1.418	1.498	1.478	1.559	1.539	1.603	1.583	1.666	1.646	1.806	1.786	16
17	1.078	1.062	1.455	1.435	1.501	1.481	1.560	1.540	1.621	1.601	1.665	1.645	1.728	1.708	1.869	1.849	17
18	1.141	1.125	1.517	1.497	1.563	1.543	1.623	1.603	1.684	1.664	1.728	1.708	1.791	1.771	1.931	1.911	18
19	1.203	1.188	1.580	1.560	1.626	1.606	1.685	1.665	1.746	1.726	1.790	1.770	1.853	1.833	1.994	1.974	19
20	1.266	1.250	1.642	1.622	1.688	1.668	1.748	1.728	1.809	1.789	1.853	1.833	1.916	1.896	2.056	2.036	20
21	1.328	1.312	1.705	1.685	1.751	1.731	1.810	1.790	1.871	1.851	1.915	1.895	1.978	1.958	2.119	2.099	21
22	1.391	1.375	1.767	1.747	1.813	1.793	1.873	1.853	1.934	1.914	1.978	1.958	2.041	2.021	2.181	2.161	22
23	1.454	1.438	1.830	1.810	1.876	1.856	1.935	1.915	1.996	1.976	2.040	2.020	2.103	2.083	2.244	2.224	23
24	1.516	1.500	1.892	1.872	1.938	1.918	1.998	1.978	2.059	2.039	2.103	2.083	2.166	2.146	2.306	2.286	24
25	1.578	1.562	1.954	1.934	2.001	1.981	2.060	2.040	2.121	2.101	2.165	2.145	2.228	2.208	2.369	2.349	25
26	1.641	1.625	2.017	1.997	2.063	2.043	2.123	2.103	2.184	2.164	2.228	2.208	2.291	2.271	2.431	2.411	26
27	1.704	1.688	2.080	2.060	2.126	2.106	2.185	2.165	2.246	2.226	2.290	2.270	2.353	2.333	2.494	2.474	27
28	1.766	1.750	2.142	2.122	2.188	2.168	2.248	2.228	2.309	2.289	2.353	2.333	2.416	2.396	2.556	2.536	28
29	1.828	1.812	2.205	2.185	2.251	2.231	2.310	2.290	2.371	2.351	2.415	2.395	2.478	2.458	2.619	2.599	29
30	1.891	1.875	2.267	2.247	2.313	2.293	2.373	2.353	2.434	2.414	2.478	2.458	2.541	2.521	2.681	2.661	30
31	1.953	1.937	2.330	2.310	2.376	2.356	2.435	2.415	2.496	2.476	2.540	2.520	2.603	2.583	2.744	2.724	31
32	2.016	2.000	2.392	2.372	2.438	2.418	2.498	2.478	2.559	2.539	2.603	2.583	2.666	2.646	2.806	2.786	32
33	2.078	2.062	2.455	2.435	2.501	2.481	2.560	2.540	2.621	2.601	2.665	2.645	2.728	2.708	2.869	2.849	33
34	2.141	2.125	2.517	2.497	2.563	2.543	2.623	2.603	2.684	2.664	2.728	2.708	2.791	2.771	2.931	2.911	34

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ARMY - AV
DLA - IS

USER SYMBOLS:

PROCUREMENT SPECIFICATION: MIL-B-23964

P.A.: NAVY - AS
Other Cust: USAF - 11
ARMY - AV

SUPERSEDES: MS3369 (AS)

TITLE: BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 63 KSI Fcu, HEXAGON SLOTTED HEAD, 450°F AND 650°F

MILITARY STANDARD

MS3369

TABLE II (CONTINUED)

GRIP DASH NO.	NOMINAL SIZE		.6250 (5/8)		.7500 (3/4)		GRIP DASH NO.
	THREAD DESIGNATION		18 UNJF - 3A		16 UNJF-3A		
	G-GRIP LENGTH		L-LENGTH		L-LENGTH		
	MAX	MIN	MAX	MIN	MAX	MIN	
03	.204	.188					03
04	.266	.250					04
05	.328	.312					05
06	.390	.375					06
07	.454	.438					07
08	.516	.500					08
09	.578	.562					09
10	.641	.625					10
11	.704	.688					11
12	.766	.750					12
13	.828	.812					13
14	.891	.875					14
15	.954	.938					15
16	1.016	1.000					16
17	1.078	1.062					17
18	1.141	1.125					18
19	1.203	1.188					19
20	1.266	1.250	2.187	2.167			20
21	1.328	1.312	2.249	2.229			21
22	1.391	1.375	2.312	2.292			22
23	1.454	1.438	2.375	2.355			23
24	1.516	1.500	2.437	2.417	2.523	2.503	24
25	1.578	1.562	2.499	2.479	2.585	2.565	25
26	1.641	1.625	2.562	2.542	2.648	2.628	26
27	1.704	1.688	2.625	2.605	2.711	2.691	27
28	1.766	1.750	2.687	2.667	2.773	2.753	28
29	1.828	1.812	2.749	2.729	2.835	2.815	29
30	1.891	1.875	2.812	2.792	2.898	2.878	30
31	1.953	1.937	2.874	2.854	2.960	2.940	31
32	2.016	2.000	2.937	2.917	3.023	3.003	32
33	2.078	2.062	2.998	2.979	3.085	3.065	33
34	2.141	2.125	3.063	3.042	3.148	3.128	34

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FED. SUP CLASS 5306

PLATE NO 22069

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USER SYMBOLS:

PROCUREMENT SPECIFICATION: MIL-B-23964

P.A.: NAVY - AS; Other Cust: USAF - 11, ARMY - AV

SUPERSEDES: MS3369 (AS)

TITLE: BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Ftu, 83 KSI Fsu, HEXAGON SLOTTED HEAD, 450° F AND 650°F

MILITARY STANDARD

MS 3369

SHEET 5 OF 8

DD FORM 672-1 (Coordinated)

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PLATE NO. 22018

TABLE III

GRIP DASH NO.	NOMINAL SIZE		.1900 (10)		.2500 (1/4)		.3125 (5/16)		.3750 (3/8)		.4375 (7/16)		.5000 (1/2)		.5625 (9/16)		GRIP DASH NO.
	THREAD DESIGNATION		32 UNJF-3A		28 UNJF-3A		24 UNJF-3A		24 UNJF-3A		20 UNJF-3A		20 UNJF-3A		18 UNJF-3A		
	G-GRIP LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		
	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	
35	2.204	2.188	2.580	2.560	2.626	2.606	2.685	2.665	2.746	2.726	2.790	2.770	2.853	2.833	2.994	2.974	35
36	2.266	2.250	2.642	2.622	2.688	2.668	2.748	2.728	2.809	2.789	2.853	2.833	2.916	2.896	3.056	3.036	36
37	2.328	2.312	2.705	2.685	2.751	2.731	2.810	2.790	2.871	2.851	2.915	2.895	2.978	2.958	3.119	3.099	37
38	2.390	2.375	2.767	2.747	2.813	2.793	2.873	2.853	2.934	2.914	2.978	2.958	3.041	3.021	3.181	3.161	38
39	2.454	2.438	2.830	2.810	2.876	2.856	2.935	2.915	2.996	2.976	3.040	3.020	3.103	3.083	3.244	3.224	39
40	2.516	2.500	2.892	2.872	2.938	2.918	2.998	2.978	3.059	3.039	3.103	3.083	3.166	3.146	3.306	3.286	40
41	2.578	2.562	2.954	2.934	3.001	2.981	3.060	3.040	3.121	3.101	3.165	3.145	3.228	3.208	3.369	3.349	41
42	2.641	2.625	3.017	2.997	3.063	3.043	3.123	3.103	3.184	3.164	3.228	3.208	3.291	3.271	3.431	3.411	42
43	2.704	2.688	3.080	3.060	3.126	3.106	3.185	3.165	3.246	3.226	3.290	3.270	3.353	3.333	3.494	3.474	43
44	2.766	2.750	3.142	3.122	3.188	3.168	3.248	3.228	3.309	3.289	3.353	3.333	3.416	3.396	3.556	3.536	44
45	2.828	2.812	3.205	3.185	3.251	3.231	3.310	3.290	3.371	3.351	3.415	3.395	3.478	3.458	3.619	3.599	45
46	2.891	2.875	3.267	3.247	3.313	3.293	3.373	3.353	3.434	3.414	3.478	3.458	3.541	3.521	3.681	3.661	46
47	2.954	2.938	3.330	3.310	3.376	3.356	3.435	3.415	3.496	3.476	3.540	3.520	3.603	3.583	3.744	3.724	47
48	3.016	3.000	3.392	3.372	3.438	3.418	3.498	3.478	3.559	3.539	3.603	3.583	3.666	3.646	3.806	3.786	48
49	3.078	3.062	3.455	3.435	3.501	3.481	3.560	3.540	3.621	3.601	3.665	3.645	3.728	3.708	3.869	3.849	49
50	3.141	3.125	3.517	3.497	3.563	3.543	3.623	3.603	3.684	3.664	3.728	3.708	3.791	3.771	3.931	3.911	50
51	3.203	3.188	3.580	3.560	3.626	3.606	3.685	3.665	3.746	3.726	3.790	3.770	3.853	3.833	3.994	3.974	51
52	3.266	3.250	3.642	3.622	3.688	3.668	3.748	3.728	3.809	3.789	3.853	3.833	3.916	3.896	4.056	4.036	52
53	3.328	3.312	3.705	3.685	3.751	3.731	3.810	3.790	3.871	3.851	3.915	3.895	3.978	3.958	4.119	4.099	53
54	3.391	3.375	3.767	3.747	3.813	3.793	3.873	3.853	3.934	3.914	3.978	3.958	4.041	4.021	4.181	4.161	54
55	3.454	3.438	3.830	3.810	3.876	3.856	3.935	3.915	3.996	3.976	4.040	4.020	4.103	4.083	4.244	4.224	55
56	3.516	3.500	3.892	3.872	3.938	3.918	3.998	3.978	4.059	4.039	4.103	4.083	4.166	4.146	4.306	4.286	56
57	3.578	3.562	3.954	3.934	4.001	3.981	4.060	4.040	4.121	4.101	4.165	4.145	4.228	4.208	4.369	4.349	57
58	3.641	3.625	4.017	3.997	4.063	4.043	4.123	4.103	4.184	4.164	4.228	4.208	4.291	4.271	4.431	4.411	58
59	3.704	3.688	4.080	4.060	4.126	4.106	4.185	4.165	4.246	4.226	4.290	4.270	4.353	4.333	4.494	4.474	59
60	3.766	3.750	4.142	4.122	4.188	4.168	4.248	4.228	4.309	4.289	4.353	4.333	4.416	4.396	4.556	4.536	60
61	3.828	3.812	4.205	4.185	4.251	4.231	4.310	4.290	4.371	4.351	4.415	4.395	4.478	4.458	4.619	4.599	61

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DD FORM 1 MAR 72 672-1 (Coordinated)

PROCUREMENT SPECIFICATION: MIL-B-23964

P.A. NAVY-AS Other Cust USAF-11 ARMY-AV

SUPERSEDES: MS3369 (AS)

TITLE: BOLTS, SELF-RETAINING, POSITIVE LOCKING CRES 90 KSI Ftu 63 KSI Fcu HEXAGON SLOTTED HEAD, 450°F AND 650°F

MILITARY STANDARD

MS3369

SHEET 6 OF 8

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO 27068

TABLE III (CONTINUED)

GRIP DASH NO.	NOMINAL SIZE		.6250 (5/8)		.7500 (3/4)		GRIP DASH NO.
	THREAD DESIGNATION		18 UNJF-3A		16 UNJF-3A		
	G-GRIP LENGTH		L-LENGTH		L-LENGTH		
	MAX	MIN	MAX	MIN	MAX	MIN	
35	2.204	2.188	3.125	3.105	3.211	3.191	35
36	2.266	2.250	3.187	3.167	3.273	3.253	36
37	2.328	2.312	3.249	3.229	3.335	3.315	37
38	2.390	2.375	3.312	3.292	3.398	3.378	38
39	2.454	2.438	3.375	3.355	3.461	3.441	39
40	2.516	2.500	3.437	3.417	3.523	3.503	40
41	2.578	2.562			3.585	3.565	41
42	2.641	2.625			3.648	3.628	42
43	2.704	2.688			3.711	3.691	43
44	2.766	2.750			3.773	3.753	44
45	2.828	2.812			3.835	3.815	45
46	2.891	2.875			3.898	3.878	46
47	2.954	2.938			3.961	3.941	47
48	3.016	3.000			4.023	4.003	48
49	3.078	3.062					
50	3.141	3.125					
51	3.203	3.188					
52	3.266	3.250					
53	3.328	3.312					
54	3.391	3.375					
55	3.454	3.438					
56	3.516	3.500					
57	3.578	3.562					
58	3.641	3.625					
59	3.704	3.688					
60	3.766	3.750					
61	3.828	3.812					

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USAF - 99
ARMY - AV
DLA - IS

USER SYMBOLS:

P.A.
NAVY - AS
Other Cust
USAF - 11
ARMY - AV

PROCUREMENT SPECIFICATION
MIL-B-22964

SUPERSEDES
MS336B (AS)

TITLE
BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 K Sl Ftu, 63 KSI Ftu, HEXAGON SLOTTED HEAD, 450° F AND 650°F

MILITARY STANDARD

MS 3369

SHEET 7 OF 8

DD FORM 1 MAR 72 672-1 (Coordinated)

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PLATE NO. 22005

FED. SUP CLASS
5306

TABLE IV

GRIP DASH NO.	NOMINAL SIZE		.1900 (10)		.2500 (1/4)		.3125 (5/16)		.3750 (3/8)		.4375 (7/16)		.5000 (1/2)		.5625 (9/16)		GRIP DASH NO.
	THREAD DESIGNATION		32 UNJF-3A		28 UNJF-3A		24 UNJF-3A		24 UNJF-3A		20 UNJF-3A		20 UNJF-3A		18 UNJF-3A		
	G-GRIP LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		
	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	
62	3.891	3.875	4.267	4.247	4.313	4.293	4.373	4.353	4.434	4.414	4.478	4.458	4.541	4.521	4.681	4.661	62
63	3.953	3.937	4.330	4.310	4.376	4.356	4.435	4.415	4.496	4.476	4.540	4.520	4.603	4.583	4.744	4.724	63
64	4.016	4.000	4.392	4.372	4.438	4.418	4.498	4.478	4.559	4.539	4.603	4.583	4.666	4.646	4.806	4.786	64
65	4.078	4.062	4.455	4.435	4.501	4.481	4.560	4.540	4.621	4.601	4.665	4.645	4.728	4.708	4.869	4.849	65
66	4.141	4.125	4.517	4.497	4.563	4.543	4.623	4.603	4.684	4.664	4.728	4.708	4.791	4.771	4.931	4.911	66
67	4.203	4.188	4.580	4.560	4.626	4.606	4.685	4.665	4.746	4.726	4.790	4.770	4.853	4.833	4.994	4.974	67
68	4.266	4.250	4.642	4.622	4.688	4.668	4.748	4.728	4.809	4.789	4.853	4.833	4.916	4.896	5.056	5.036	68
69	4.328	4.312	4.705	4.685	4.751	4.731	4.810	4.790	4.871	4.851	4.915	4.895	4.978	4.958	5.119	5.099	69
70	4.391	4.375	4.767	4.747	4.813	4.793	4.873	4.853	4.934	4.914	4.978	4.958	5.041	5.021	5.181	5.161	70
71	4.454	4.438	4.830	4.810	4.876	4.856	4.935	4.915	4.996	4.976	5.040	5.020	5.103	5.083	5.244	5.224	71
72	4.516	4.500	4.892	4.872	4.938	4.918	4.998	4.978	5.059	5.039	5.103	5.083	5.166	5.146	5.306	5.286	72
73	4.578	4.562	4.954	4.934	5.001	4.981	5.060	5.040	5.121	5.101	5.165	5.145	5.228	5.208	5.369	5.349	73
74	4.641	4.625	5.017	4.997	5.063	5.043	5.123	5.103	5.184	5.164	5.228	5.208	5.291	5.271	5.431	5.411	74
75	4.704	4.688	5.080	5.060	5.126	5.106	5.185	5.165	5.246	5.226	5.290	5.270	5.353	5.333	5.494	5.474	75
76	4.766	4.750	5.142	5.122	5.188	5.168	5.248	5.228	5.309	5.289	5.353	5.333	5.416	5.396	5.556	5.536	76
77	4.828	4.812	5.205	5.185	5.251	5.231	5.310	5.290	5.371	5.351	5.415	5.395	5.478	5.458	5.619	5.599	77
78	4.891	4.875	5.267	5.247	5.313	5.293	5.373	5.353	5.434	5.414	5.478	5.458	5.541	5.521	5.681	5.661	78
79	4.953	4.937	5.330	5.310	5.376	5.356	5.435	5.415	5.496	5.476	5.540	5.520	5.603	5.583	5.744	5.724	79
80	5.016	5.000	5.392	5.372	5.438	5.418	5.498	5.478	5.559	5.539	5.603	5.583	5.666	5.646	5.806	5.786	80
81	5.078	5.062	5.455	5.435	5.501	5.481	5.560	5.540	5.621	5.601	5.665	5.645	5.728	5.708	5.869	5.849	81
82	5.141	5.125	5.517	5.497	5.563	5.543	5.623	5.603	5.684	5.664	5.728	5.708	5.791	5.771	5.931	5.911	82
83	5.203	5.188	5.580	5.560	5.626	5.606	5.685	5.665	5.746	5.726	5.790	5.770	5.853	5.833	5.994	5.974	83
84	5.266	5.250	5.642	5.622	5.688	5.668	5.748	5.728	5.809	5.789	5.853	5.833	5.916	5.896	6.056	6.036	84
85	5.328	5.312	5.705	5.685	5.751	5.731	5.810	5.790	5.871	5.851	5.915	5.895	5.978	5.958	6.119	6.099	85
86	5.390	5.375	5.767	5.747	5.813	5.793	5.873	5.853	5.934	5.914	5.978	5.958	6.041	6.021	6.181	6.161	86
87	5.454	5.438			5.876	5.856	5.935	5.915									
96	6.016	6.000			6.438	6.418											
97	6.078	6.062			6.501	6.481											
111	6.953	6.937			7.376	7.356											

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TABLE V

SIZE	SHANK DOUBLE SHEAR ULTIMATE STRENGTH LBS MIN	BOLT AND NUT ASSEMBLY ULTIMATE TENSILE STRENGTH LBS MIN SEE NOTE 13	RETAINING DEVICE ULTIMATE TENSILE LOAD LBS MIN
.1900	5,150	1,110	110
.2500	9,200	2,040	230
.3125	14,400	3,250	510
.3750	20,600	5,050	575
.4375	28,000	6,800	710
.5000	36,800	9,250	1,160
.5625	46,000	11,800	1,420
.6250	57,500	15,050	2,070
.7500	82,500	22,000	2,950

NOTES:

1. MATERIALS: BOLT AND PLUNGER: CORROSION RESISTING STEEL PH 13-8 Mo CHEMISTRY OF AMS 5629
   BALLS OR ELEMENT: CORROSION RESISTING STEEL PER QQ-S-763 CLASS FSC 400C
   SPRING: CORROSION RESISTING STEEL WIRE, SPRING TEMPER PER AMS 5673, 17-7 PH OR QQ-W-423, CLASS FS302, CONDITION B
2. HEAT TREATMENT: BOLT AND PLUNGER: PER MIL-H-6875 TO CONDITION H1000
   BALLS: PER MIL-H-6875 TO ROCKWELL C50 TO C62
3. MATERIALS, NONMAGNETIC: BOLT AND PLUNGER: CORROSION RESISTING STEEL A-286 CHEMISTRY OF AMS 5737
   BALLS OR ELEMENT: CORROSION RESISTING STEEL PER MIL-S-7720
   SPRING: NICKEL-CHROMIUM-IRON ALLOY PER QQ-W-390 CONDITION C OR NICKEL-ALLOY SPRING PER AMS 5699, SPRING TEMPER
4. HEAT TREATMENT: BALLS: PER MIL-H-6875 TO ROCKWELL C25 MINIMUM
5. PROTECTIVE TREATMENT: CRES MATERIAL PASSIVATED PER QQ-P-35
6. SURFACE TEXTURE: UNTHREADED SHANK AND BEARING SURFACE UNDER HEAD SHALL BE 32 MICROINCHES, ALL OTHER SURFACES SHALL BE 125 MICROINCHES IN ACCORDANCE WITH ANSI B46.1
7. BOLTS SHALL BE FREE FROM BURRS AND SLIVERS.
8. EXAMPLE OF PART NUMBER:

   MS3369A 416 = .2500 UNJF-3A BOLT, 1 INCH NOMINAL GRIP, A-286 MATERIAL

   - 416: SEE TABLES I, II, III AND IV FOR DASH NUMBERS
   - GRIP DASH NUMBER
   - DIAMETER DASH NUMBER
   - A: MATERIAL - "A" FOR A-286, "B" FOR PH 13-8Mo, "C" FOR PH 17-4
   - MS NUMBER
9. DESIGN USAGE: FOR DESIGN AND USAGE LIMITATIONS, SEE MS33602.
10. MAGNETIC PERMEABILITY: FOR BOLT ASSEMBLY MADE FROM A-286 MATERIAL MAGNETIC PERMEABILITY SHALL BE LESS THAN 2.0 (AIR-1.0) FOR A FIELD STRENGTH = 200 ORSTEADS USING MAGNETIC PERMEABILITY INDICATOR PER MIL-I-17214 or EQUIVALENT.
11. TEMPERATURE LIMITATION: 450°F MAX. FOR PH 13-8Mo AND 650°F MAX. FOR A-286 MATERIAL.
12. INTERCHANGEABILITY RELATIONSHIP: BOLTS MADE FROM PH 17-4 (CODE "-") ARE INACTIVE FOR NEW DESIGN AND CANCELLED AFTER 15 MAY 1981.
    BOLTS MADE FROM CRES PH 13-8 Mo (CODE "B") MAY BE SUBSTITUTED FOR BOLTS MADE FORM PH 17-4 IN ALL APPLICATIONS.
    BOLTS MADE FROM PH 17-4 (CODED "-") SHALL NOT BE USED IN PLACE OF BOLTS MADE FROM CRES PH 13-8 Mo.
    BOLTS MADE FROM CRES PH 17-4 SHALL NOT BE USED AFTER 1 JAN 1983.
13. TENSILE LOAD IS BASED ON THE AREA AT THREAD MINOR DIAMETER IN FED-STD-H28/2 AND 62.5 KSI Ftu.
14. BOLT PLUNGER MANUFACTURED FROM PH 13-8Mo AND A-286 MATERIAL SHALL BE COATED WITH DRY FILM LUBRICANT PER MIL-L-8937 AND MIL-L-46010 RESPECTIVELY.
15. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.
16. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

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P.A. NAVY - AS Other Cust: USAF - 11, ARMY - AV	TITLE: BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 63 KSI Ftu, HEXAGON SLOTTED HEAD, 450°F AND 650°F	MILITARY STANDARD MS3369
PROCUREMENT SPECIFICATION: MIL-B-23964	SUPERSEDES: MS3369(AS)	SHEET 8 OF 8

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